Build the fan controls one state at a time

Task 4.6 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Label motor terminals and sketch the teacher-checked switch paths. Show where polarity reverses and what changes speed. Disconnect power before rewiring.

List the required speeds and directions in a state table. Mark the intended polarity at the motor for each direction.

State	Switch settings	Expected polarity	Predicted motion
Off			
Forward low			
Forward high			
Reverse low			
Reverse high			

02 Plan it on paper

Draw a circuit for each switch state. Trace the current path and identify the component changing the speed.

My prediction, chosen feature or design limit

Build the fan controls one state at a time

Task 4.6 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Use the teacher-checked diagram for the available kit. Build one state at a time. Verify direction and two distinct speed settings before combining the controls. Ask a partner to predict the motor state from the diagram before operating it. Record differences and disconnect power before correcting a connection.

State tested	Actual direction / speed	Mismatch or fix
Forward settings		
Reverse settings		
Off		

04 Use the reading

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Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Which control changes direction and which changes speed in your circuit? Support it with an observation.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.